HTML5 and CSS3

A Refresher

Overview

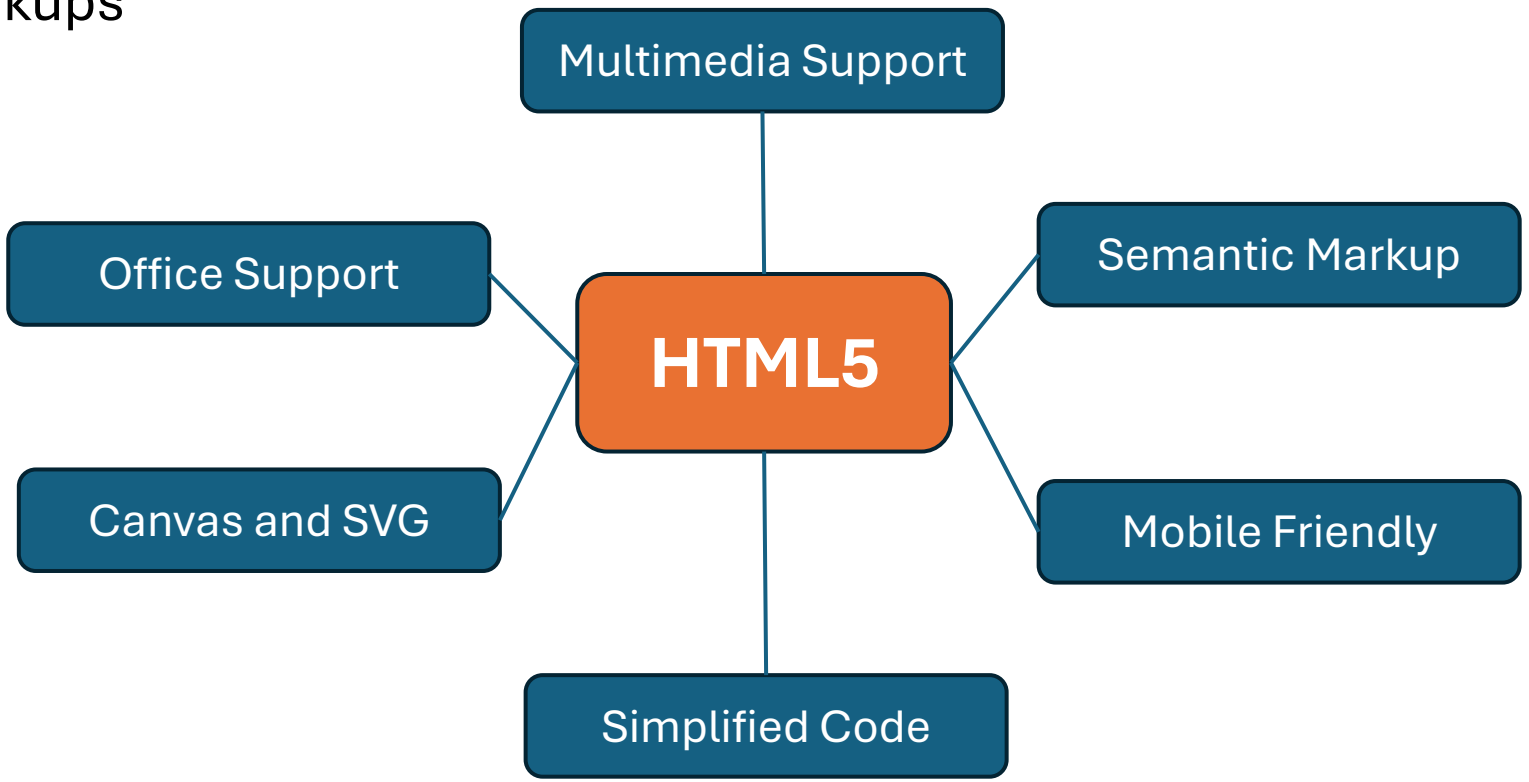
- Discuss HTML5 features and semantic elements
- Understand the HTML5 codes
- Discuss the CSS3 features
- Understand the CSS3 codes

What is HTML?

- HTML is defined as **HyperText Markup Language**
- It was developed by Timothy “Tim” Berners-Lee in 1991
- Original HTML had 20 tags and displayed simple text with basic markup for formatting
- HTML 2 added many of the attributes known today such as images, forms, input
- HTML 3 added new tags that have been part of minor revisions
- HTML 4 extended the number of tags. Currently most HTML is 4.01.

HTML5 New Features

- Semantic elements and Markups
- Audio and video support
- Canvas
- Drag and drop
- Local data storage
- Offline applications
- Server events
- Geolocation



HTML5 Semantic Elements

A semantic element clearly describes its meaning to both the browser and the developer.

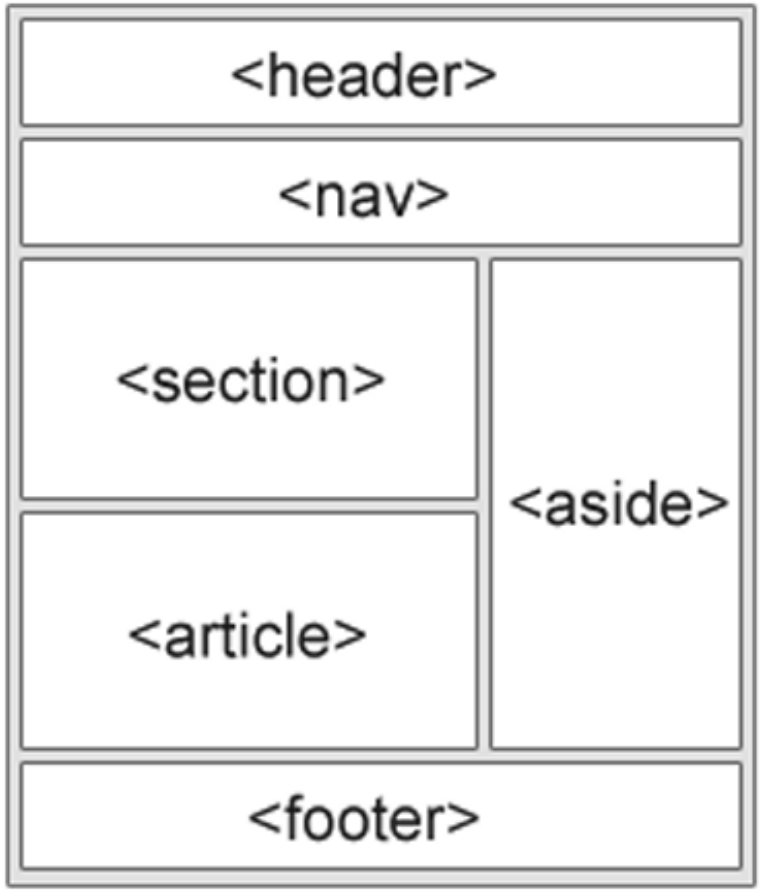
Examples of non-semantic elements: and - Tells nothing about its content.

Examples of semantic elements: <form>, <table>, and <img> - Clearly defines its content.

Semantic Element: some other

<header>
<section>
<aside>
<nav>
<article>
<figure>

<figcaption>
<footer>
<details>
<summary>
<mark>
<time>



Audio and Video

Audio

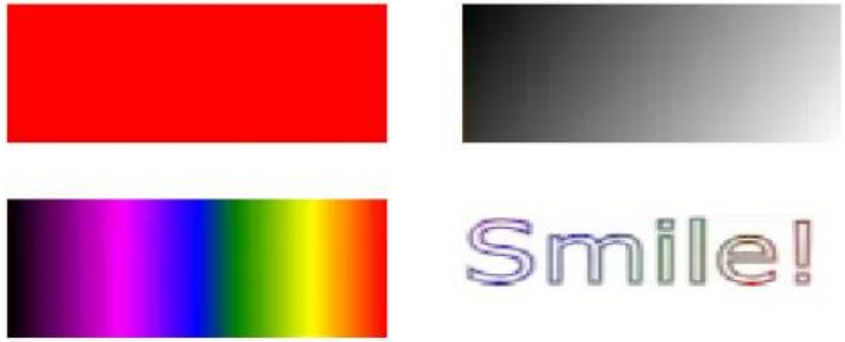
```
<audio controls>
 <source src="horse.ogg" type="audio/ogg">
 <source src="horse.mp3" type="audio/mpeg">
Your browser does not support the audio element.
</audio>
```

Video

```
<video width="320" height="240" controls>
 <source src="movie.mp4" type="video/mp4">
 <source src="movie.ogg" type="video/ogg">
Your browser does not support the video tag.
</video>
```

Canvas

- HMTL **<canvas>** element is used to draw graphics, on the fly, via scripting (usually JavaScript).
- Canvas has several methods of drawing paths, boxes, circles, text, and adding images.
- It is only a container for graphics. You must use a script to actually draw the graphics.



Server Events

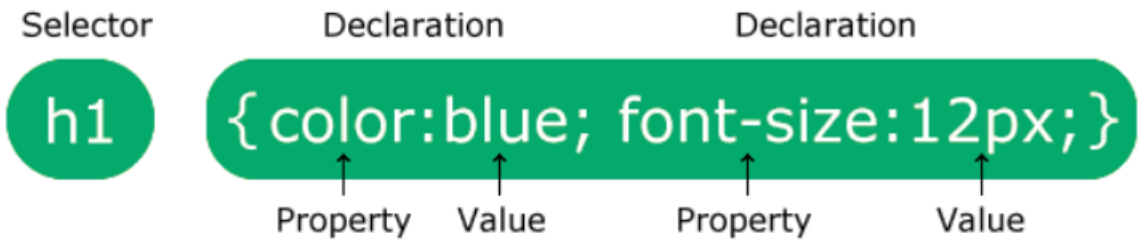
- A server-sent event is when a web page automatically gets updates from a server.
- Before, the web page would have to ask if any updates were available. With server-sent events, the updates come automatically.
- Examples: Facebook/Twitter updates, stock price updates, news feeds, sport results, etc.

Geolocation

- The HTML5 Geolocation API is used to get the geographical position of a user.
- Since this can compromise user privacy, the position is not available unless the user approves it.

What is CSS?

- Cascading Style Sheet
- CSS 1 was proposed in 1995 as a way to style documents. It contained descriptions for font, background and other style tag.
- CSS 2 is the version we currently use adding positioning and other elements.
- A CSS rule consists of a selector and a declaration block:



CSS New Styles

- border-radius – Round Corners
- border-colors – Gradient Borders
- box-shadow – Drop Shadows
- text-Shadow – Text Drop Shadows
- gradient – Gradient backgrounds
- resize – Resize an element
- background-size – resize background
- background – support multiple images

Benefits of HTML5 and CSS3

- Rich interface elements not requiring third party plugins
- Separation of content and design
- Enhanced mobile support
- New programming interfaces

Sample Code - HTML

```
<html>
<head><title>Integrative Programming</title></head>
<body>
    <p> This is a sample web page created in HTML</p>
</body>
</html>
```

Sample Code - CSS

```
<style>
    p {
      color: red;
      text-align: center;
    }
</style>
```

Note: *The <style> tag is placed in the <head> </head> tag.*

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